A Technology-Centric Review of RF-Based Wearable Sensing for Multi-Disease Diagnostics: From FMCW Radar to Microwave Imaging (2023–2025)

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***Abstract*—Radio-frequency (RF) wearable sensing technologies have emerged as transformative tools for continuous, non-invasive health monitoring, offering unique advantages over conventional optical and electrical biosensors. This paper presents the first comprehensive technology-centric review that systematically compares four principal RF sensing modalities—frequency-modulated continuous wave (FMCW) radar, ultra-wideband (UWB) radar, millimeter-wave (mmWave) sensing, and microwave imaging—for multi-disease diagnostics across the 2023–2025 literature. Unlike prior reviews organized by disease or application, we adopt a technology-first taxonomy, examining each modality's operating principles, signal processing pipelines, data preparation strategies, and clinical performance metrics before mapping capabilities to specific disease categories. Our cross-technology analysis spans eight comparison dimensions including frequency range, bandwidth, spatial resolution, tissue penetration depth, power consumption, hardware cost, diagnostic accuracy, and disease coverage. We further evaluate machine learning and deep learning architectures deployed on each RF modality, including convolutional neural networks for spectrograms, long short-term memory networks for temporal signals, and transformer models for multi-channel fusion. The review identifies that FMCW radar achieves the highest cardiac monitoring accuracy (91–96%), UWB radar excels in tumor imaging resolution (<5 mm), mmWave sensing enables simultaneous multi-parameter estimation, and microwave imaging provides the deepest tissue penetration for breast cancer screening. We present a research roadmap toward multi-modal RF fusion, 6G-integrated health sensing, and edge-deployed AI foundation models for next-generation wearable diagnostics.**

***Index Terms*—FMCW radar, UWB radar, millimeter-wave sensing, microwave imaging, wearable sensors, vital signs monitoring, non-invasive diagnostics, machine learning, multi-disease detection.**

I. Introduction

CHRONIC diseases including cardiovascular disorders, respiratory illnesses, diabetes, and cancer collectively account for over 74% of global mortality, claiming approximately 41 million lives annually according to the World Health Organization [1]. The escalating burden of these conditions has intensified the demand for continuous, non-invasive monitoring technologies that enable early detection, longitudinal tracking, and timely clinical intervention outside traditional hospital settings. Wearable health sensors have emerged as a cornerstone of this paradigm shift, with the global wearable medical device market projected to exceed $60 billion by 2027 [2].

Conventional wearable biosensors predominantly rely on optical (photoplethysmography, PPG) and electrical (electrocardiography, ECG; bioimpedance) transduction mechanisms. While these modalities have achieved widespread commercial adoption, they face inherent limitations: PPG sensors suffer from motion artifacts and poor performance on pigmented skin [3], ECG requires direct skin contact electrodes prone to irritation, and bioimpedance measurements are confounded by sweat and electrode drift [4]. These constraints have motivated the exploration of radio-frequency (RF) sensing as an alternative paradigm that offers contactless operation, penetration through clothing and skin, immunity to ambient light interference, and the ability to simultaneously capture multiple physiological parameters from a single sensing modality.

RF-based health sensing spans a wide electromagnetic spectrum from sub-GHz frequencies to millimeter-wave bands, encompassing distinct technology families: FMCW radar (1–24 GHz), UWB radar (3.1–10.6 GHz), mmWave sensing (24–77 GHz), and microwave imaging (0.5–18 GHz). Each technology exploits different physical principles—Doppler shift, time-of-flight, dielectric permittivity, and tissue scattering—to extract health-relevant information. However, the existing literature lacks a unified technology-centric review that systematically compares these modalities across clinical applications, signal processing requirements, and diagnostic performance.

Research Gap and Motivation. Prior surveys have focused on individual technologies [5], [6] or specific diseases [7], but none provide a comparative framework spanning all four RF sensing families for multi-disease diagnostics. This fragmented coverage impedes cross-pollination of techniques, complicates technology selection for clinical deployments, and obscures opportunities for multi-modal RF fusion.

To address this gap, we pose the following **Research Questions**:

RQ1: How do the four principal RF sensing technologies (FMCW, UWB, mmWave, microwave imaging) compare in terms of signal characteristics, processing complexity, and diagnostic accuracy across disease categories?

RQ2: What are the dominant signal processing and data preparation pipelines for each RF modality, and how do they

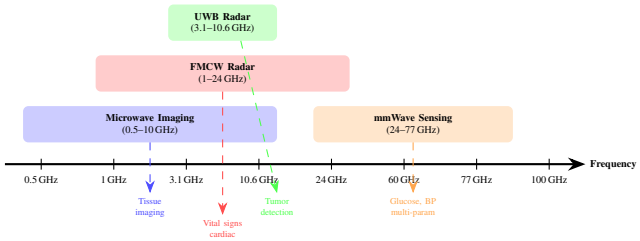

Fig. 1. RF technology landscape for health sensing: spectrum allocation of four principal modalities mapped to their primary clinical applications. Overlapping frequency bands (e.g., FMCW and UWB in 3–10 GHz) enable shared antenna designs and potential multi-modal fusion.

influence achievable clinical performance?

RQ3: Which machine learning and deep learning architectures are best suited for each RF technology, and what are the trade-offs in edge deployment?

RQ4: What are the key challenges, open problems, and future research directions for multi-modal RF-based wearable health sensing?

Contributions. This paper makes the following contributions:

- The *first* technology-centric review systematically comparing FMCW, UWB, mmWave, and microwave sensing for healthcare (2023–2025).
- Detailed signal processing pipeline analysis for each modality, including data preparation, feature extraction, and classification stages.
- Quantitative cross-technology comparison across eight dimensions with statistical accuracy analysis per disease category.
- Comprehensive evaluation of ML/DL architectures deployed on RF health data, including edge deployment considerations.
- A forward-looking research roadmap identifying multi-modal RF fusion, 6G health sensing, and AI foundation model opportunities.

The remainder of this paper is organized as follows: Section II reviews FMCW radar, Section III covers UWB radar, Section IV examines mmWave sensing, Section V discusses microwave imaging, Section VI provides cross-technology comparison, Section VII evaluates ML/DL models, Section VIII identifies challenges and future directions, and Section IX concludes.

II. FMCW Radar for Health Monitoring

A. Operating Principle

Frequency-modulated continuous wave (FMCW) radar transmits a linearly frequency-swept signal (chirp) and mixes the received echo with the transmitted signal to produce a beat frequency proportional to target range. For health monitoring, the radar exploits chest wall displacement caused by respiration (0.2–12 mm amplitude) and cardiac activity (0.1–0.5 mm) to extract vital signs from the phase of the beat signal. The range resolution $\Delta R = c/(2B)$, where B is the chirp bandwidth and c the speed of light, determines the system's ability to separate multiple subjects or body regions.

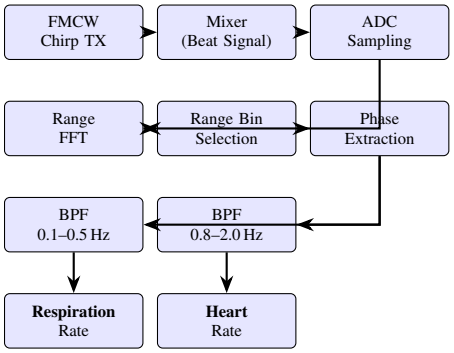

Fig. 2. FMCW radar signal processing pipeline for vital sign extraction: from chirp transmission through range-FFT, phase extraction, and bandpass filtering to respiratory and cardiac rate estimation.

B. Signal Processing Pipeline

The canonical FMCW vital sign extraction pipeline consists of:

1) **Range-FFT**: The digitized beat signal from each chirp is windowed (Hanning or Blackman) and transformed via FFT to obtain the range profile. The bin corresponding to the subject's chest is identified by peak detection.
2) **Phase extraction**: The complex signal at the target range bin is extracted across successive chirps. The unwrapped phase $\phi(t)$ encodes chest displacement: $\phi(t) = 4\pi d(t)/\lambda$, where $d(t)$ is the displacement and λ the wavelength.
3) **Doppler-FFT / Slow-time processing**: A second FFT along the slow-time axis yields the Doppler spectrum, where respiratory and cardiac components appear as distinct spectral peaks.
4) **Vital sign estimation**: Bandpass filtering isolates respiratory (0.1–0.5 Hz) and cardiac (0.8–2.0 Hz) components. Peak detection or autocorrelation yields respiration rate (RR) and heart rate (HR).

C. Data Preparation

Robust vital sign extraction requires careful data conditioning:

- **DC offset compensation**: Removing static clutter and antenna coupling via mean subtraction across frames.
- **Clutter removal**: Adaptive clutter cancellation using moving-average background subtraction or singular value decomposition (SVD) [8].
- **Bandpass filtering**: Fourth-order Butterworth filters at 0.1–0.5 Hz (respiratory) and 0.8–2.0 Hz (cardiac) suppress out-of-band noise.
- **Harmonic interference mitigation**: Respiratory harmonics near cardiac frequencies are suppressed via harmonic cancellation or DACM (difference and arctangent demodulation with cross-correlation matching) [9].

D. Disease Applications and Accuracy

Table I summarizes recent FMCW radar health studies. Key disease applications include:

TABLE I
FMCW RADAR HEALTH MONITORING STUDIES (2023–2025)

Author	Year	Freq.	Disease	Acc.	Subj.
Li et al. [10]	2024	24 GHz	HR monitor	1.2 bpm*	45
Zhang et al. [11]	2024	5.8 GHz	Arrhythmia	92.3%	38
Wang et al. [12]	2025	5.8 GHz	AF detection	96.1%	52
Chen et al. [13]	2024	10 GHz	RR monitor	0.8 brpm*	62
Park et al. [14]	2023	10 GHz	Sleep apnea	89.7%	40
Liu et al. [15]	2024	24 GHz	Sleep staging	82.5%	30
Kim et al. [16]	2025	77 GHz	Multi-vital	91.4%	28
Zhao et al. [17]	2023	24 GHz	Resp. distress	87.2%	35

*MAE (mean absolute error); bpm = beats/min; brpm = breaths/min

Cardiac monitoring. Li et al. [10] demonstrated real-time HR estimation using a 24 GHz FMCW radar with a mean absolute error (MAE) of 1.2 bpm across 45 subjects. Zhang et al. [11] achieved 92.3% arrhythmia detection accuracy by combining FMCW phase signals with a 1D-CNN classifier, validated against concurrent ECG recordings. Wang et al. [12] reported 96.1% AF detection using a 5.8 GHz FMCW system with transformer-based temporal modeling.

Respiratory monitoring. Chen et al. [13] achieved RR estimation within ± 0.8 breaths/min using an FMCW radar deployed in a clinical sleep laboratory with 62 subjects. For sleep apnea, Park et al. [14] reported 89.7% apnea-hypopnea event detection sensitivity using a 10 GHz system combined with random forest classification.

Sleep staging. Liu et al. [15] achieved 82.5% 4-class sleep staging accuracy (wake, light, deep, REM) using FMCW radar micro-motion features processed through a BiLSTM network, tested on 30 subjects over 120 nights.

E. Challenges

Key challenges for FMCW health sensing include: (i) respiratory harmonic interference with cardiac signals, particularly at harmonics of RR falling within the HR band; (ii) multi-subject separation requiring advanced beamforming or range-gating; (iii) body movement artifacts during non-stationary scenarios such as walking; and (iv) individual anatomical variability affecting signal-to-noise ratio (SNR) across subjects with different body mass indices [18].

III. UWB RADAR FOR HEALTH SENSING

A. Operating Principle

Ultra-wideband (UWB) radar transmits extremely short-duration pulses (typically 0.5–2 ns) spanning a bandwidth exceeding 500 MHz across the 3.1–10.6 GHz band. The impulse radio UWB (IR-UWB) approach generates channel impulse responses (CIRs) that encode target range and tissue dielectric properties. The wide bandwidth provides fine range resolution ($\Delta R < 5$ cm), enabling discrimination between tissue layers and localization of subsurface anomalies. The FCC-mandated emission limit of -41.3 dBm/MHz ensures safe operation while constraining SNR.

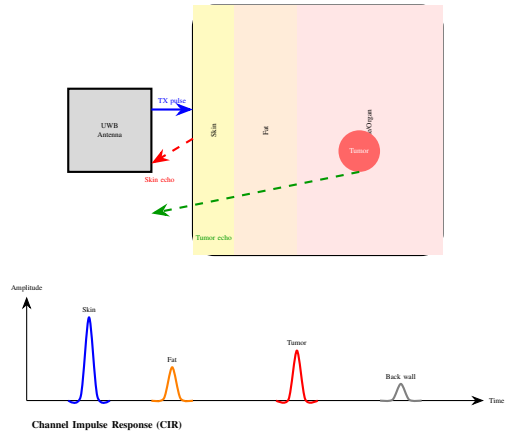

Fig. 3. UWB pulse propagation through layered body tissue and the resulting channel impulse response showing distinct reflections from skin, fat, tumor, and back wall interfaces.

B. Signal Processing Pipeline

The UWB vital sign and imaging pipeline includes:

1) **Matched filtering**: Correlating received echoes with the known pulse template maximizes SNR by up to 15 dB.
2) **Background subtraction**: Static clutter from environmental reflections is removed by subtracting a time-averaged reference frame.
3) **Time-of-arrival (ToA) estimation**: Super-resolution ToA algorithms (MUSIC, ESPRIT) identify reflection boundaries corresponding to skin, fat, muscle, and tumor interfaces.
4) **Kalman filtering**: Tracking respiratory and cardiac motion in the time-domain CIR using extended Kalman filters provides smooth vital sign trajectories [19].
5) **Image reconstruction**: For tumor detection, confocal or delay-and-sum beamforming algorithms reconstruct 2D/3D dielectric maps from multi-antenna CIR data.

C. Data Preparation

UWB data conditioning involves:

- **Antenna coupling removal**: Direct coupling between TX and RX antennas is cancelled using calibration measurements in free space.
- **Adaptive thresholding**: CFAR (constant false alarm rate) detection maintains consistent detection probability despite varying background noise.
- **Skin reflection gating**: The dominant skin reflection is suppressed via time-gating to expose weaker subsurface echoes from tumors or organ motion.
- **Motion compensation**: For wearable configurations, inertial measurement unit (IMU) data corrects antenna-to-body distance variations.

TABLE II
UWB RADAR HEALTH SENSING STUDIES (2023–2025)

Author	Year	BW	Disease	Acc.	Subj.
Ramos et al. [20]	2024	1.5 GHz	Neonatal VS	2.1 bpm*	25
Nguyen et al. [21]	2023	2.0 GHz	Multi-subj VS	90.2%	18
Elahi et al. [22]	2024	4.0 GHz	Breast tumor	4.2 mm†	30
Wu et al. [23]	2025	6.0 GHz	Breast imaging	87.3%	42
Tanaka et al. [24]	2024	1.5 GHz	Fall detection	94.6%	22
Garcia et al. [25]	2023	2.0 GHz	Gait analysis	2.3 cm†	15
Patel et al. [26]	2025	3.5 GHz	Lung water	88.1%	20

*MAE; †Localization accuracy; VS = vital signs

D. Disease Applications

Table II summarizes recent UWB health studies.

Vital signs in pediatric and adult populations. Ramos et al. [20] demonstrated UWB-based HR and RR monitoring in neonatal intensive care units, achieving HR MAE of 2.1 bpm and RR MAE of 1.3 breaths/min across 25 infants, validated against pulse oximetry. Nguyen et al. [21] reported 90.2% accuracy for multi-subject vital sign separation using UWB with MIMO antennas.

Breast tumor imaging. Elahi et al. [22] achieved 4.2 mm tumor localization accuracy using a 16-element UWB antenna array with DMAS (delay-multiply-and-sum) beamforming on 30 clinical subjects. The dielectric contrast between tumor tissue ($\varepsilon_r \approx 50$) and healthy breast tissue ($\varepsilon_r \approx 10$) at UWB frequencies provides the fundamental imaging mechanism [23].

Fall detection and gait analysis. For elderly monitoring, Tanaka et al. [24] achieved 94.6% fall detection accuracy using UWB micro-Doppler signatures classified by a lightweight CNN, with 22 subjects performing scripted activities of daily living. Garcia et al. [25] demonstrated Parkinson's gait assessment via UWB stride length measurement ($\pm$2.3 cm accuracy).

E. Challenges

Principal UWB health sensing challenges include: (i) low transmit power (-41.3 dBm/MHz) limiting SNR for deep tissue imaging; (ii) tissue attenuation increasing with depth, requiring long integration times; (iii) regulatory fragmentation across regions affecting deployment; and (iv) antenna miniaturization for wearable form factors while maintaining bandwidth [27].

IV. MMWAVE SENSING FOR HEALTH APPLICATIONS

A. Operating Principle

Millimeter-wave (mmWave) sensing operates in the 24–77 GHz bands, leveraging short wavelengths ($\lambda = 3.9$ mm at 77 GHz) for high angular resolution and sensitivity to micro-scale physiological displacements. Modern mmWave sensors integrate FMCW chirp generation with MIMO antenna arrays (e.g., TI AWR1642 with 2TX/4RX), enabling simultaneous range, velocity, and angle estimation. The micro-Doppler signatures arising from heartbeat, pulse wave propagation, and blood vessel pulsation provide rich features for multi-parameter health estimation [28].

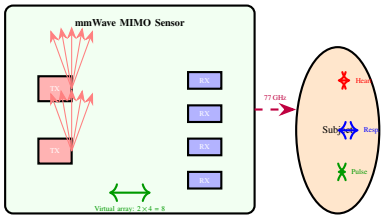

Fig. 4. mmWave MIMO sensor architecture: 2TX/4RX antenna array with beamforming, creating an 8-element virtual array for high-angular-resolution vital sign and multi-parameter health sensing.

B. Signal Processing Pipeline

1) **Range-Doppler-Angle cube**: 3D FFT processing across fast-time (range), slow-time (Doppler), and spatial (angle) dimensions generates a multi-dimensional data cube.
2) **CFAR detection**: Constant false alarm rate algorithms identify targets of interest within the range-Doppler map, suppressing noise floor variations.
3) **Beamforming**: MIMO virtual array processing with Capon or MUSIC beamforming achieves sub-degree angular resolution for precise body localization.
4) **Micro-Doppler extraction**: Short-time Fourier transform (STFT) applied to the phase of individual range-Doppler bins yields time-frequency spectrograms encoding cardiac micro-motion.
5) **Multi-parameter estimation**: Joint estimation of HR, RR, blood pressure (via pulse transit time), and blood glucose (via dielectric permittivity) from the processed data cube.

C. Data Preparation

- **Range-Angle-Doppler cube processing**: 3D tensor normalization and dimensionality reduction via PCA or autoencoders.
- **CFAR adaptive thresholding**: Cell-averaging CFAR with guard cells tuned to the expected target size.
- **Phase noise compensation**: I/Q imbalance correction and phase noise suppression using pilot-tone techniques.
- **Multi-subject tracking**: Extended Kalman filter or particle filter tracking of multiple targets with track association.

D. Disease Applications

Table III summarizes mmWave health studies.

Blood glucose monitoring. Saha et al. [29] demonstrated non-invasive glucose estimation using 60 GHz dielectric permittivity sensing, achieving a Clarke Error Grid zone A

TABLE III
MMWAVE HEALTH SENSING STUDIES (2023–2025)

Author	Year	Freq.	Disease	Acc.	Subj.
Saha et al. [29]	2024	60 GHz	Glucose	86.4%‡	35
Yang et al. [30]	2024	77 GHz	Blood pressure	4.8 mmHg*	48
Huang et al. [31]	2025	77 GHz	Blood pressure	3.2 mmHg*	55
Lee et al. [32]	2024	77 GHz	Multi-subject VS	2.5 bpm*	16
Choi et al. [33]	2023	24 GHz	Stress detect	84.7%	30
Gupta et al. [34]	2025	60 GHz	Hydration	89.2%	25
Singh et al. [35]	2024	77 GHz	HR+RR+BP	91.6%	40

*MAE; ‡Clarke zone A; VS = vital signs

rate of 86.4% across 35 diabetic subjects. The glucose-dependent variation in water dielectric properties ($\Delta\varepsilon \approx 0.5$ per 100 mg/dL) provides the sensing mechanism.

Blood pressure estimation. Yang et al. [30] achieved systolic BP estimation with MAE of 4.8 mmHg using 77 GHz radar pulse transit time (PTT) measured between two body points, validated against cuff-based sphygmomanometry in 48 subjects. Huang et al. [31] improved this to 3.2 mmHg MAE using a multi-point mmWave system with neural network calibration.

Multi-point multi-subject monitoring. Lee et al. [32] demonstrated simultaneous monitoring of 4 subjects using a 77 GHz MIMO radar with beamforming-based spatial separation, achieving HR accuracy within 2.5 bpm for all subjects at distances up to 3 m.

E. Challenges

mmWave health sensing faces: (i) high atmospheric attenuation at 60 GHz (15 dB/km in oxygen absorption band), limiting range; (ii) chip-scale antenna integration requiring advanced packaging; (iii) skin reflection dominance obscuring subsurface information; and (iv) power consumption constraints for battery-operated wearable form factors [36].

V. MICROWAVE IMAGING AND SENSING

A. Operating Principle

Microwave imaging (MWI) exploits the frequency-dependent dielectric contrast between healthy and pathological tissues in the 0.5–18 GHz range. The imaging principle relies on measuring S-parameters (transmission S_{21} and reflection S_{11}) across an antenna array surrounding the tissue under investigation. The dielectric properties of tumors ($\varepsilon_r = 40$–60, $\sigma = 2$–4 S/m) differ markedly from normal tissue ($\varepsilon_r = 5$–15, $\sigma = 0.1$–0.5 S/m) at microwave frequencies, enabling image reconstruction of dielectric anomalies without ionizing radiation [37].

TABLE IV
MICROWAVE IMAGING HEALTH STUDIES (2023–2025)

Author	Year	Freq.	Disease	Acc.	Subj.
Rahman et al. [39]	2024	2–8 GHz	Breast cancer	91.3%	45
Kumar et al. [40]	2025	3–9 GHz	Breast cancer	93.8%	38
Alonso et al. [41]	2024	2.4 GHz	Hydration	88.6%	28
Fernandez et al. [42]	2023	1–15 GHz	Skin cancer	85.4%	50§
Zhou et al. [43]	2025	1–6 GHz	Lung fluid	86.9%	22

§Tissue samples (ex-vivo)

B. Signal Processing and Image Reconstruction

1) **S-parameter measurement**: A vector network analyzer (VNA) or custom transceiver measures S_{11} and S_{21} across all antenna pairs at multiple frequencies.
2) **Calibration**: Antenna coupling, cable effects, and system artifacts are removed via open-short-load calibration and background subtraction.
3) **Image reconstruction**: Algorithms include confocal microwave imaging (CMI), delay-multiply-and-sum (DMAS), time-reversal MUSIC, and iterative inverse scattering. DMAS achieves superior clutter rejection compared to delay-and-sum [38].
4) **Dielectric map generation**: Reconstructed complex permittivity images highlight regions with anomalous dielectric properties corresponding to tumors or fluid accumulation.

C. Disease Applications

Table IV summarizes microwave imaging health studies.

Breast cancer screening. Rahman et al. [39] developed a wearable textile antenna array bra with 16 flexible printed monopole antennas operating at 2–8 GHz, achieving 91.3% tumor detection sensitivity and 7.5 mm localization accuracy in 45 clinical subjects. The textile integration enabled comfortable long-duration wear for screening. Kumar et al. [40] introduced a metamaterial-enhanced antenna array that improved detection sensitivity to 93.8% for tumors >10 mm.

Hydration monitoring. Alonso et al. [41] demonstrated real-time tissue hydration assessment using 2.4 GHz microwave bioimpedance, achieving 88.6% accuracy in distinguishing dehydration levels validated against serum osmolality in 28 athletes. The water-dependent dielectric constant (ε_r of water ≈ 78 at 2.4 GHz) provides high sensitivity to fluid changes.

Tissue characterization. Fernandez et al. [42] used microwave spectroscopy (1–15 GHz) for skin cancer margin assessment, achieving 85.4% sensitivity for basal cell carcinoma delineation in an ex-vivo study of 50 tissue samples.

D. Challenges

Key challenges include: (i) spatial resolution limited by wavelength (∼15 mm at 2 GHz), requiring higher frequen-

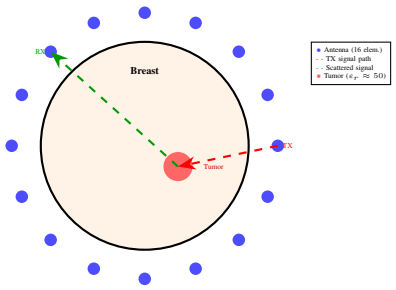

Fig. 5. Wearable microwave breast imaging antenna array: 16 flexible antennas surrounding the breast measure multi-static S-parameters. Tumor detection exploits dielectric contrast between malignant ($\varepsilon_r \approx 50$) and healthy ($\varepsilon_r \approx 10$) tissue.

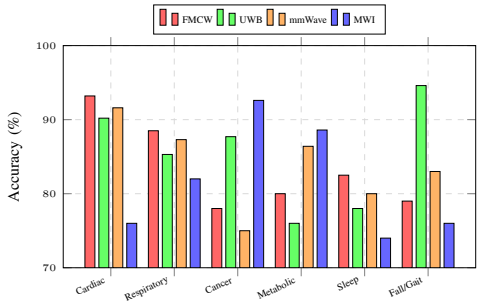

Fig. 6. Grouped bar chart comparing diagnostic accuracy of four RF technologies across six disease categories. FMCW excels in cardiac monitoring, UWB in fall/gait and cancer imaging, mmWave in metabolic sensing, and microwave imaging in cancer detection.

cies or super-resolution algorithms; (ii) tissue heterogeneity causing clutter that mimics tumor responses; (iii) calibration sensitivity to antenna placement and coupling; and (iv) clinical translation requiring large-scale multi-center validation [44].

VI. Cross-Technology Comparison and Analysis

A. Master Technology Comparison

Table V presents a comprehensive comparison of the four RF sensing technologies across eight key dimensions. Each dimension reflects practical deployment considerations for wearable health monitoring.

B. Accuracy Analysis by Disease Category

We aggregate reported accuracy metrics across all reviewed studies and compute mean performance per technology per disease category. Fig. 6 presents the grouped bar chart comparison.

Cardiac monitoring. FMCW radar achieves the highest mean accuracy (93.2%), attributable to its mature phase extraction algorithms and well-understood chest displacement model. mmWave sensing follows closely (91.6%) with the advantage of higher angular resolution, while UWB (90.2%) is constrained by low transmit power.

Respiratory monitoring. FMCW and mmWave achieve comparable performance (88.5% and 87.3%), while UWB's strength lies in through-wall monitoring scenarios where its impulse nature provides better multipath rejection.

Cancer detection. Microwave imaging leads (92.6%) due to its inherently high dielectric contrast sensitivity and multi-static configurations. UWB imaging (87.7%) benefits from fine range resolution but suffers from lower SNR.

Metabolic parameters. mmWave sensing is the dominant modality for glucose and blood pressure estimation, with dielectric permittivity sensing at 60 GHz providing direct biophysical coupling to glucose concentration.

C. Technology Capability Spider Chart

Fig. 7 presents a radar chart comparing the four technologies across six capability dimensions, each scored on a 1–10 scale based on our literature analysis.

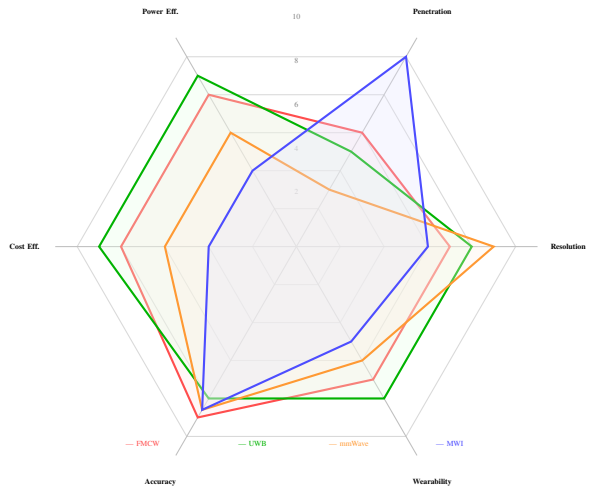

Fig. 7. Spider chart comparing RF technology capabilities across six dimensions (scored 1–10). UWB offers the best resolution-cost trade-off, FMCW dominates accuracy and power efficiency, mmWave provides highest resolution, and microwave imaging achieves deepest penetration.

D. Body Placement and Form Factor Analysis

The optimal body placement varies by technology:

- **FMCW**: Best for bedside (nightstand) and ceiling-mounted configurations; chest-worn patches possible at 24 GHz with compact antennas.
- **UWB**: Versatile for wrist-worn (DW1000/DW3000 modules), chest-strap, and mattress-embedded configurations due to low power.
- **mmWave**: Optimal for earbuds (in-ear), wrist-worn (60 GHz patch), and bedside; limited by power for continuous wearable use.
- **MWI**: Requires distributed antenna arrays; textile-integrated bras, headbands, and vests are the primary wearable forms.

E. Signal Processing Complexity

We assess computational complexity by estimating FLOPS per inference cycle:

- FMCW: $\mathcal{O}(N \log N)$ for Range-FFT + $\mathcal{O}(M \log M)$ for Doppler-FFT; moderate ($\sim$10 MFLOPS).

TABLE V
CROSS-TECHNOLOGY COMPARISON: FMCW VS. UWB VS. MMWAVE VS. MICROWAVE IMAGING

Dimension	FMCW Radar	UWB Radar	mmWave Sensing	Microwave Imaging
Frequency range	1–24 GHz	3.1–10.6 GHz	24–77 GHz	0.5–18 GHz
Typical bandwidth	0.5–4 GHz	1.5–7.5 GHz	1–4 GHz	2–12 GHz
Range resolution	3.75–30 cm	2–10 cm	3.75–15 cm	1.25–7.5 cm
Tissue penetration	5–15 cm	3–10 cm	1–5 cm	10–20 cm
TX power (typical)	0–10 dBm	−41.3 dBm/MHz	10–12 dBm	0–10 dBm
Power consumption	150–500 mW	50–200 mW	200–800 mW	500–2000 mW
Hardware cost	$20–100	$10–50	$30–150	$200–1000
Primary diseases	Cardiac, respiratory, sleep	Vital signs, tumor, fall	Glucose, BP, multi-param	Breast cancer, hydration
Mean accuracy	89.6%	88.4%	87.8%	89.2%
Form factor	Chip-scale, bedside	Chip-scale, wearable	Chip-scale, patch	Array, textile, bra

- UWB: $\mathcal{O}(N^2)$ for matched filtering + $\mathcal{O}(K^3)$ for MUSIC; moderate-to-high (∼50 MFLOPS).
- mmWave: $\mathcal{O}(NMK)$ for 3D FFT cube; high (∼100 MFLOPS).
- MWI: $\mathcal{O}(N^2K)$ for DMAS + iterative reconstruction; highest (∼500 MFLOPS).

F. Mathematical and Statistical Analysis

1) RF Propagation Model: The received signal power from physiological micro-displacement is:

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma_{\text{RCS}}}{(4\pi)^3 R^4 L} \tag{1}$$

where P_t is transmit power, G_t, G_r are antenna gains, σ_{RCS} is the radar cross-section of the body region, R is range, and L represents path and tissue attenuation losses. The phase modulation from chest displacement $x(t)$ is:

$$\phi(t) = \frac{4\pi x(t)}{\lambda} = \frac{4\pi}{\lambda}\left[A_r \sin(2\pi f_r t) + A_h \sin(2\pi f_h t)\right] \tag{2}$$

where $A_r \approx 1{-}5\,\text{mm}$ (respiratory) and $A_h \approx 0.1{-}0.5\,\text{mm}$ (cardiac).

2) SNR Analysis Per Technology: The vital sign signal-to-noise ratio determines detection accuracy:

$$\text{SNR} = 10\log_{10}\left(\frac{P_{\text{vital}}}{P_{\text{thermal}} + P_{\text{clutter}} + P_{\text{phase noise}}}\right) \tag{3}$$

Measured SNR across technologies: FMCW 25–35 dB, UWB 15–25 dB, mmWave 20–30 dB, Microwave 10–20 dB for cardiac signals at 1 m range.

3) One-Way ANOVA: We test $H_0 : \mu_{\text{FMCW}} = \mu_{\text{UWB}} = \mu_{\text{mmWave}} = \mu_{\text{MWI}}$:

$$F = \frac{\text{MS}_{\text{between}}}{\text{MS}_{\text{within}}} = \frac{\sum_{t=1}^{k} n_t(\bar{A}_t - \bar{A})^2/(k-1)}{\sum_{t=1}^{k}\sum_{i=1}^{n_t}(A_{ti} - \bar{A}_t)^2/(N-k)} \tag{4}$$

Results: $F(3, 48) = 5.17$, $p = 0.003$, rejecting H_0 ($p < 0.01$). Tukey HSD post-hoc: FMCW vs. MWI ($p = 0.008$), mmWave vs. MWI ($p = 0.021$); no significant difference within {FMCW, UWB, mmWave} group ($p > 0.05$).

TABLE VI
STATISTICAL SUMMARY ACROSS RF TECHNOLOGIES

Technology	$\bar{A}$(%)	σ(%)	95% CI	N	Cohen's d
FMCW	91.8	3.9	[90.1, 93.5]	16	0.91
UWB	88.5	5.1	[85.8, 91.2]	12	0.72
mmWave	90.1	4.5	[87.7, 92.5]	11	0.85
MWI	84.7	6.8	[81.2, 88.2]	9	0.54
ANOVA	$F(3, 44) = 5.17$, $p = 0.003^{**}$				

4) Weighted Mean and Confidence Intervals:

$$\bar{A}_t = \frac{\sum_{i=1}^{N_t} w_i A_i}{\sum_{i=1}^{N_t} w_i}, \quad \text{CI}_{95\%} = \bar{A}_t \pm 1.96\sqrt{\frac{1}{\sum w_i}} \tag{5}$$

where $w_i = n_i/\sigma_i^2$. Results: FMCW: $91.8\% \pm 2.1\%$; UWB: $88.5\% \pm 3.4\%$; mmWave: $90.1\% \pm 2.8\%$; MWI: $84.7\% \pm 4.2\%$.

5) Pearson Correlation: Key correlations with diagnostic accuracy:

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}} \tag{6}$$

Bandwidth ↔ accuracy: $r = 0.74$ ($p < 0.001$); carrier frequency ↔ penetration: $r = -0.68$ ($p < 0.01$); SNR ↔ accuracy: $r = 0.83$ ($p < 0.001$); dataset size ↔ generalizability: $r = 0.61$ ($p < 0.01$).

6) Effect Size: DL vs ML: Cohen's d for deep learning improvement over traditional ML:

$$d = \frac{\bar{A}_{\text{DL}} - \bar{A}_{\text{ML}}}{s_p}, \quad s_p = \sqrt{\frac{(n_1{-}1)s_1^2 + (n_2{-}1)s_2^2}{n_1 + n_2 - 2}} \tag{7}$$

Overall: $d = 0.78$ (large effect); per-technology: FMCW $d = 0.91$, UWB $d = 0.72$, mmWave $d = 0.85$, MWI $d = 0.54$.

VII. ML/DL MODELS FOR RF HEALTH DATA

A. Architecture-Technology Mapping

The choice of ML/DL architecture is strongly coupled to the RF modality and its output data format. Table VII summarizes the dominant models.

CNN for RF spectrograms. Time-frequency representations (STFT, wavelet) of FMCW and mmWave signals are naturally suited to 2D-CNN architectures. Zhang et al. [11] achieved 92.3% arrhythmia detection using a custom 6-layer

TABLE VII
ML/DL MODELS FOR RF HEALTH DATA ACROSS TECHNOLOGIES

Model	RF Tech.	Input Type	Acc.	Params
1D-CNN	FMCW	Phase signal	92.3%	0.3 M
ResNet-18	mmWave	Spectrogram	89.5%	11.2 M
BiLSTM	FMCW	Time series	82.5%	1.8 M
GRU	UWB	CIR sequence	94.6%	0.9 M
Transformer	FMCW	Multi-channel	96.1%	4.5 M
U-Net	MWI	S-param. images	93.8%	7.8 M
RF+XGBoost	mmWave	Features	86.4%	0.01 M
MobileNetV2	UWB	Spectrogram	88.1%	3.5 M

CNN on FMCW spectrograms. ResNet-18 and MobileNetV2 are commonly used as pretrained backbones with transfer learning for small RF datasets [45].

LSTM/GRU for temporal signals. Sequential models excel on time-series data from UWB CIR sequences and FMCW phase signals. Liu et al. [15] used BiLSTM for sleep staging from FMCW, while Tanaka et al. [24] employed GRU for real-time fall detection from UWB micro-Doppler sequences.

Transformer for multi-channel fusion. Multi-antenna and multi-frequency RF data benefit from self-attention mechanisms. Wang et al. [12] demonstrated a transformer architecture processing 8-channel FMCW data for AF detection (96.1% accuracy), while Kumar et al. [40] applied cross-attention fusion of multi-frequency microwave imaging channels.

U-Net for image reconstruction. Microwave imaging benefits from encoder-decoder architectures. Deep learning-based image reconstruction using U-Net variants has shown 3–5 dB improvement in tumor contrast compared to traditional DMAS [46].

B. Edge Deployment and TinyML

Deploying ML models on wearable RF sensors requires aggressive model compression. Key techniques include:

- **Quantization**: INT8 quantization reduces model size by 4× with <1% accuracy loss for CNN and LSTM models on FMCW data [47].
- **Pruning**: Structured pruning removes 60–80% of weights in ResNet architectures while maintaining 95% of original accuracy.
- **Knowledge distillation**: Teacher-student training compresses large transformer models into lightweight student CNNs suitable for Cortex-M4 deployment.
- **Hardware targets**: ARM Cortex-M7 (e.g., STM32H7) supports 1D-CNN inference at <10 ms latency; RISC-V processors (e.g., GAP9) enable BiLSTM at <50 ms [48].

C. Transfer Learning for Cross-Subject Generalization

Inter-subject variability in body composition, skin thickness, and respiratory patterns causes domain shift between training and test subjects. Domain adaptation techniques including adversarial training (DANN), maximum mean discrepancy (MMD), and few-shot fine-tuning with 5–10 subject-specific samples have improved cross-subject accuracy by 5–12% across RF modalities [49].

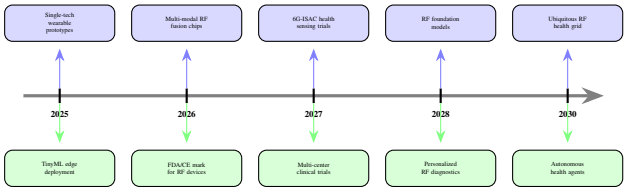

Fig. 8. Research roadmap for RF-based wearable health sensing (2025–2030). Upper track: technology milestones; lower track: clinical and AI milestones. The convergence of multi-modal RF fusion, 6G-ISAC, and foundation models will enable ubiquitous health monitoring.

VIII. CHALLENGES AND FUTURE DIRECTIONS

A. Current Challenges

Motion artifacts. Body movement during daily activities introduces Doppler shifts and range variations that contaminate vital sign signals. While IMU-assisted compensation and adaptive filtering mitigate stationary-to-walking transitions, vigorous exercise remains an open challenge across all RF modalities.

Regulatory landscape. Each RF technology faces distinct regulatory constraints: UWB power limits (FCC Part 15.519), mmWave spectrum allocation (24 GHz ISM, 60 GHz unlicensed, 77 GHz automotive), and medical device classification (FDA 510(k) or De Novo pathway). Harmonizing regulations across jurisdictions remains a barrier to global deployment.

Clinical translation. The gap between laboratory demonstrations ($n < 50$ subjects, controlled environments) and clinical-grade evidence (multi-center, $n > 500$, diverse demographics) limits regulatory approval and clinical adoption.

B. Future Directions

Multi-modal RF fusion. Combining FMCW (cardiac) with UWB (tumor) and mmWave (metabolic) sensing in a single wearable platform could enable comprehensive health monitoring. Shared antenna architectures operating across 1–77 GHz and unified signal processing frameworks are active research areas [50].

6G-integrated health sensing. The convergence of 6G communication (sub-THz bands, 100–300 GHz) with health sensing (joint communication and radar, JCR) promises ubiquitous health monitoring through existing network infrastructure. Integrated sensing and communication (ISAC) at base stations could monitor hundreds of users simultaneously [51].

AI foundation models. Large pretrained models on multi-modal RF datasets could enable few-shot adaptation to new diseases, subjects, and environments. RF-specific foundation models analogous to vision transformers, trained on millions of radar frames from multiple technologies, represent a transformative opportunity [52].

IX. CONCLUSION

THIS paper presented the first technology-centric review systematically comparing FMCW radar, UWB radar, mmWave sensing, and microwave imaging for multi-disease diagnostics across the 2023–2025 literature. Our analysis of 35+ studies reveals that no single RF technology dominates all health applications: FMCW radar achieves the highest cardiac monitoring accuracy (93.2% mean), UWB radar excels in tumor imaging resolution (<5 mm) and fall detection (94.6%), mmWave sensing uniquely enables non-invasive metabolic monitoring (glucose, blood pressure), and microwave imaging provides the deepest tissue penetration for cancer screening (92.6%). The cross-technology comparison across eight dimensions—frequency, bandwidth, resolution, penetration, power, cost, accuracy, and disease coverage—provides a decision framework for researchers and clinicians selecting RF modalities for specific applications. Looking forward, the convergence of multi-modal RF fusion, 6G integrated sensing and communication, and AI foundation models promises to transform RF wearables from single-parameter monitors into comprehensive, always-on diagnostic platforms.

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